University of Amsterdam

Replicating a classic with AI assistants

Card and Krueger (1994): minimum wages and fast-food employment

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[date] · [room]

AI & Research · Lab 4, evening 3 · 60 minutes, follow along in pairs

Plan

Minutes	What happens	You do
0–5	My area: optimization, and how I use agents	Listen
5–10	The question, the paper, the design	Vote, then listen
10–30	Round 1: replicate Table 4	Follow along, in pairs
30–42	Round 2: stress-test one choice	Pairs, one task each
42–47	An assistant explains the paper	Watch and listen
47–60	What no software settles	Discuss

Homework: explain the paper to the public, and a one-page report.

My area: optimization

How I work with agents

Read

I ask for a slide deck that tells me a paper's main story.

Code and test

Every number comes from a run, with checks that can fail.

Write

Agents draft articles, slides and emails; I rewrite them.

Replicate

Agents rebuild published results from the authors' data and code.

GitHub

Version control and pull requests, also to open-source repositories.

Stay critical

I question every iteration, and compare frontier models: Codex and Claude.

What agents change for me

Not more papers, but more ambitious ones

Agents do not turn me into a paper mill. They let me take on research of a larger scope, especially where coding is a substantial part of the work.

A critical reviewer of the code

I have written code by hand for thirty years, in many languages. That makes me very critical of how agents develop code: I read it, test it and push back.

A textbook with Krzysztof Postek, Alessandro Zocca and Jeffrey Kantor (Cambridge University Press, 2025). Every model is a notebook you can run, check and change: https://mobook.github.io/MO-book/

Demo: an agent runs one of the book's notebooks copy and paste

> Run the notebook 02-production-planning-basic.ipynb and tell me the optimal plan and profit. Then: the demand for product U is capped at 40 units. What happens if that cap rises to 60? Change only that in a copy, run it, and show both results side by side.

	$U \leq 40$	$U \leq 60$
Profit	2600	2600
Units of U, V	20, 60	20, 60
Labour A, B used	80, 100	80, 100

Nothing changes. The agent noticed the surprise and checked why: the cap on U was never binding; labour A and B are used up (shadow prices 70 and 50, the U cap 0).

With agents: replicating a location benchmark

The problem

Brooklyn: choose 50 of 7,730 sites to cover as many of 2,504,700 people as possible within 800 m.

The published work

Máximo, Cordeau and Nascimento (2025), with Java code.

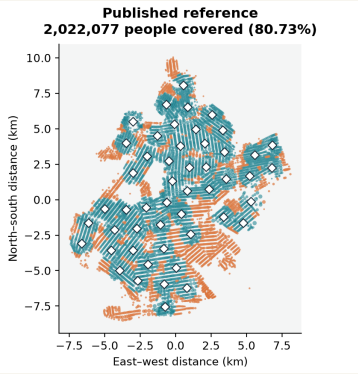
https://doi.org/10.1111/itor.13387

https://github.com/vinymax10/AILS-MCLP

What the agent did

Compiled the authors' code, ran it for 60 seconds, checked the coverage.

It recovered exactly the 50 published sites.



Teal: covered Orange: not covered Diamonds: chosen sites

The papers contain no maps of their solutions.

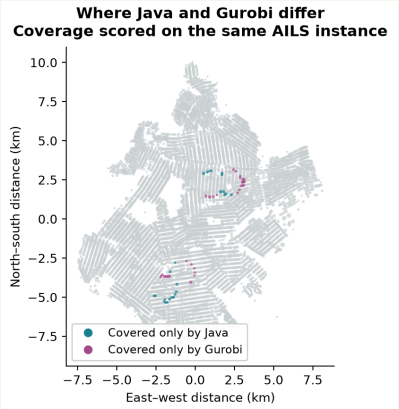
These maps are agentic work too.

We seemed to beat the literature

Solution	People covered
Our Gurobi sites, our own scoring	2,024,813
Published reference	2,022,077
Authors' Java code, one run	2,022,077
Our Gurobi sites, scored their way	2,021,487

Our "win" came from a slightly different coverage rule: who counts as within 800 m. Scored the same way, we cover 590 people *fewer*.

Replication depends on the model's details, not only on the algorithm.



Map made by the agent, not taken from the papers.

Optimization or econometrics:
the agent gives you a number in minutes.

Knowing why it is right is still your job.

Let's try that together, on a famous paper.

A longer example with agents: https://github.com/gromicho/OceanCleanupDataChallenge

✋ Vote with your hand

New Jersey raises the minimum wage from $4.25 to $5.05.
Pennsylvania, next door, does not.
What happens to fast-food employment, relative to Pennsylvania?

A. it falls B. no change C. it rises

The paper

A natural experiment

- 1 April 1992: New Jersey's minimum wage rises from \$4.25 to \$5.05.
- Pennsylvania, next door, keeps \$4.25: the comparison group.
- Telephone survey of 410 fast-food restaurants: Burger King, KFC, Roy Rogers, Wendy's.
- Wave 1 in February–March 1992, wave 2 seven to eight months later.

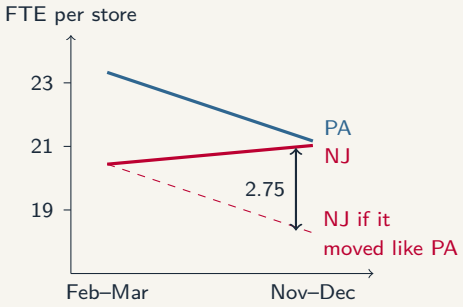
Everything is public

Paper: https://davidcard.berkeley.edu/papers/min-wage-ff-nj.pdf

Data, codebook and the authors' SAS program:
https://davidcard.berkeley.edu/data_sets/njmin.zip

SAS is commercial statistics software, standard in 1990s economics. You do not need it: the agent reads `check.sas` and rewrites it in Python.

Difference in differences



	Before	After	Change
PA	23.33	21.17	−2.16
NJ	20.44	21.03	+0.59
Difference in changes			+2.75

FTE: full-time equivalent, part-timers count half.

Our computation; the paper prints 2.76.

The regressions in Table 4

$$\Delta E_i = a + b\,X_i + c\,NJ_i + \varepsilon_i \qquad \text{or} \qquad \Delta E_i = a' + b'X_i + c'\,GAP_i + \varepsilon_i'$$

- ΔE_i: change in FTE employment at store i.
- X_i: chain and company-ownership dummies; in model (v) also region dummies.
- GAP_i: proportional wage rise store i needs to reach \$5.05; zero in Pennsylvania and for New Jersey stores already at \$5.05.

Published	(i)	(ii)	(iii)	(iv)	(v)
NJ dummy	2.33 (1.19)	2.30 (1.20)			
GAP			15.65 (6.08)	14.92 (6.21)	11.91 (7.39)

357 stores. This is your target.

Hands-on

Round 1: replicate Table 4

20 minutes **follow along: I type it, you type it**

1. Replicate copy and paste

Download into this folder https://davidcard.berkeley.edu/papers/min-wage-ff-nj.pdf and https://davidcard.berkeley.edu/data_sets/njmin.zip and unzip it.
Read Table 4 in the paper. Translate check.sas into Python with pandas and NumPy, reproduce Table 4, and print each coefficient and standard error next to the value in the paper.

Claude Code asks before it downloads: allow it. Download blocked? Save both files in the folder yourself.

2. Then ask copy and paste

Which stores were left out of the regressions, and why?

Done when: 357 stores, all five coefficients and standard errors match. **You need:** Claude Code (Pro), Python with pandas and NumPy, an empty folder.

Checkpoint: what the agent returned

Hands up: all five match? three or four? still debugging?

A fresh agent, given only the Round 1 prompt. Lines copied from its output.

```
Stores in public.dat: 410; in DATA C1 subset: 357; used in PROC REG (no missing in any model var): 357  [paper: 357]

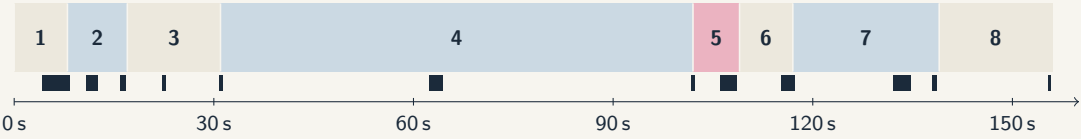
Col    Regressor      Coef    Paper       SE    Paper     SER   Paper  p(ctrl)   Paper
(i)    NJ             2.33     2.33     1.19     1.19    8.79    8.79        -       -
(ii)   NJ             2.30     2.30     1.20     1.20    8.78    8.78     0.34    0.34
(iii)  GAP           15.65    15.65     6.08     6.08    8.76    8.76        -       -
(iv)   GAP           14.92    14.92     6.21     6.21    8.76    8.76     0.44    0.44
(v)    GAP           11.91    11.91     7.39     7.39    8.74    8.75     0.31    0.40
(v)*   GAP           11.98    11.91     7.42     7.39    8.75    8.75     0.40    0.40
```

Tell the room: what did your assistant get wrong first? Did it, too, flag column (v)?

Unasked, the agent flagged column (v); its line (v)* adds ownership.

Dry run: 2.6 minutes, step by step

Bars: what the agent was doing



Black marks below: commands actually running

1. Download paper and data
2. Unzip; read code, codebook, Table 4
3. Check the data layout; start a log
4. Write the Python translation (71 s)
5. First run: all values match, (v) flagged
6. Follow-up: which stores were dropped
7. Save output and log
8. Write its report

What the agent wrote: SAS into Python

check.sas (1997)

```
EMPTOT=EMPPT*.5 + EMPFT + NMGRS;



DATA C1;
SET CHECK;
IF DEMP NE . ;
IF CLOSED=1 OR (CLOSED=0 AND DWAGE NE .);



MODEL DEMP=GAP BK KFC ROYS
  CENTRALJ SOUTHJ PA1 PA2;
```

replicate_table4.py (the agent)

```
d["emptot"] = d.emppt * 0.5 + d.empft + d.nmgrs

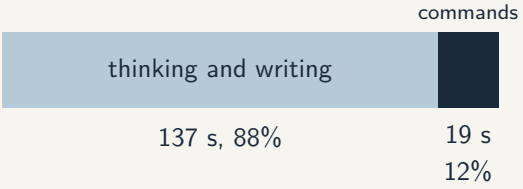


# DATA C1: IF DEMP NE .; IF CLOSED=1 OR (CLOSED=0 AND DWAGE NE .);
c1 = d[d.demp.notna()]
c1 = c1[(c1.closed == 1) | ((c1.closed == 0) & c1.dwage.notna())]
```

```
"gap+chain+region": ["gap", "bk", "kfc", "roys", "centralj",
                     "southj", "pa1", "pa2"],
# Variant: column (v) as the paper's notes describe it
VARIANT = ["gap", "bk", "kfc", "roys", "co_owned", ...]
```

Look closer: the paper's values are typed in from the PDF image; matches are checked by eye, nothing fails on a mismatch; and to obey "pandas and NumPy" it hand-coded the F-test p-value (30 lines).

Dry run: where the time went



- Downloading, unzipping and running the code took seconds.
- Almost all the time is the agent reading, reasoning and writing; nearly half of it on the script.
- The first run matched; nothing had to be fixed.

Expect longer in the room

Claude Code asks permission before downloads and commands, Python has to be found, and models and speeds differ. Round 1 has 20 minutes for that reason.

Round 2: stress-test one choice

12 minutes **one letter per pair; report one number and one sentence** copy and paste

A	Re-run model (i) on all 384 stores with employment data. Which stores does Table 4 leave out, and how did their employment change?
B	Re-run all five models with robust (HC1) standard errors. Is the New Jersey effect still significant at 5%?
C	Re-run models (i) and (iii) counting part-time workers as 0, 0.4, 0.6 and 1 instead of 0.5.
D	Re-run model (i) without the six closed stores. Then set the four temporarily closed stores to zero instead.
E	Add company ownership to model (v), as the Table 4 footnote says. Compare with the code.

Round 2, advanced cards: the econometrics behind Table 5

optional **for pairs who finish early, or as homework** copy and paste

F Is a model in changes right? Add wave-1 employment to model (iv) and instrument it with the number of cash registers (NREGS) and registers open at 11:00 (NREGS11). Compare with plain OLS.

G Reproduce Table 5 of the paper, rows 1 to 9 and 12. Print each value next to the paper's, and say which cells you cannot match and why.

F is an instrumental-variables test from the paper's text (p. 784). **G** is a partial replication: some rows are defined only vaguely in the paper.

Round 2: what we found

A Sample	384 → 357 moves the NJ estimate from 2.75 to 2.33. The 27 dropped stores *work against* the finding: 19 NJ stores gained 4.55 FTE, 8 PA stores lost 3.59.
B Standard errors	NJ dummy: conventional 1.19 ($t \approx 1.96$), robust 1.45 ($t \approx 1.61$).
C Part-timers	Weight 0 to 1: NJ from 2.55 to 2.10; GAP from 18.16 to 13.15.
D Closures	Drop the six: 2.28 (1.19). Temporary closures as zero: 2.51 (1.18), 388 stores.
E Footnote vs code	Code 11.91 (7.39); with ownership 11.98 (7.42). And there is more: see the next slide.

Compare with the board first. Our reference run, one machine, one tool.

Why replicate: one footnote, two regressions

The paper says

Footnote b of Table 4: model (v) includes the chain dummies *and* company ownership.

The code does

`check.sas` runs model (v) with the chain and region dummies, *without* ownership.

The table shows

Coefficient 11.91 (7.39): the code's regression.

SER 8.75 and $p = 0.40$: the footnote's regression.

The point

Reading the paper, you would never notice. Only rebuilding every number shows that one column mixes two regressions. Small here, but this is how you learn what a paper actually did.

Advanced cards: what we found

F: instrumental variables

Coefficient of wave-1 FTE	
Paper, IV	0.04 (0.24)
Ours, IV (348 stores)	0.03 (0.23)
Ours, plain OLS	−0.56 (0.04)

Plain OLS is pulled down by measurement error; the instrument removes that. A model in changes is fine.

G: Table 5

21 of 36 published cells match at two decimals.

Match: weighting by initial size, the Pennsylvania placebo, most call-back and closure rows.

No match: interview-week dummies; percentage columns of the FTE variants.

Paper vs paper: Table 5 prints 2.30 (1.19) where Table 4 prints 2.30 (1.20).

The pattern

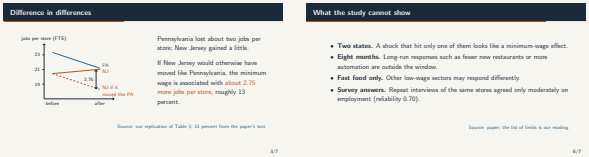
Robustness checks with a clear definition reproduce; the vaguely described ones do not.

An assistant explains the paper

With the replication done, the same assistant wrote three explanations for the public:

- an eight-slide deck
- a 600-word newspaper article
- a podcast, here cut to one minute

Listen for: what does it simplify, and is that still true?



Two slides of the example deck. "Why it mattered" relies on the assistant's background knowledge, not on the paper: it needs checking.

Discussion

Only two states.

Any shock that hits New Jersey but not Pennsylvania looks exactly like a minimum-wage effect.

What evidence would convince you?

Homework: explain it to the public

Pick one format. Paste its prompt into the assistant that holds your replication. copy and paste

Slide deck

Using the paper and our results, write an HTML slide deck of at most eight slides explaining this study to first-year students. Say where each number comes from.

Newspaper article

Using the paper and our results, write a 600-word newspaper article about this study. Invent no quotes. Include what it cannot show.

Podcast script

Using the paper and our results, write a five-minute podcast script for a host and an economist. End with the problem of having only two states.

Optional, to hear the podcast:

Turn the script into one audio file with two of my computer's built-in voices. Install nothing.

Homework: fact-check it, and report

Fact-check your explanation

1. Does every number appear in the paper or in your results?
2. Which sentence claims more than the study shows?
3. What important limit is missing?
4. Is any quote, source or name invented?

One-page report

- your prompts, in order
- the modelling choices you made or accepted
- your checks, and how they could have failed
- what you still do not trust

Exit question, tonight

Which single choice would you report first in a referee report on this paper, and why?

Take-aways

- An assistant reproduces the published numbers in minutes; knowing why they are right takes the rest of the session.
- The useful findings are in the choices: sample, errors, weights, closures, code vs footnote.
- Ask a second tool or person; disagreement is information.
- An assistant writes a fluent explanation in minutes; every number and claim in it still needs checking.
- Reproduction is not causal proof: the design carries the argument.

Paper: https://davidcard.berkeley.edu/papers/min-wage-ff-nj.pdf
Data and program: https://davidcard.berkeley.edu/data_sets/njmin.zip

Not yet tested

The podcast as audio on a Mac

We have not tested whether the optional audio prompt produces a two-voice file on a Mac.

On Windows we did it ourselves for the one-minute version, with the built-in voices. Joining two voices into one file is easy on Windows. On a Mac it may take Claude Code a little more work.

If you try it on a Mac, please tell us how it went.